An exact counterexample to cosine-sum positivity

MathDB 369172

Principia Math In Progress

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Task 1. Resolve the positivity conjecture

For an integer $d \geq 2$, choose the square-root branch equal to 1 at the origin and define

$$\sqrt{1+z+\cdots+z^d} = \sum_{k\geq 0} b_k^{(d)} z^k, \qquad \mathcal{T}_n^{(d)}(t) = \sum_{k=0}^{n} b_k^{(d)} \cos(kt).$$

Kovalev's Conjecture 1.3 [1] asserts that these cosine sums are positive for every $d \geq 2$, $n \geq 0$ and real t. It is false.

Theorem 1. *Let $d = 43$, $n = 1800$ and $t = \arccos(99/100)$. Then*

$$-\frac{61}{10000} < \mathcal{T}_{1800}^{(43)}(t) < -\frac{6099}{1000000} < 0.$$

Proof. All parts of the counterexample can be checked with integers. Put $q = d + 1 = 44$ and

$$f(z) = \sqrt{\frac{1-z^q}{1-z}} = \sum_{k\geq 0} b_k z^k.$$

Logarithmic differentiation gives

$$2(1-z)(1-z^q)f'(z) = \bigl(1 - qz^{q-1} + (q-1)z^q\bigr) f(z).$$

Comparing coefficients, with negative-index coefficients set to zero, yields

$$2kb_k = (2k-1)b_{k-1} + (2k-3q)b_{k-q} + (3q+1-2k)b_{k-q-1}.$$

Consequently the scaled coefficients $B_k = 4^k b_k$ satisfy $B_0 = 1$ and

$$\begin{aligned} 2kB_k = {} & 4(2k-1)B_{k-1} + 4^{44}(2k-132)B_{k-44} \\ & + 4^{45}(133-2k)B_{k-45}. \end{aligned} \tag{1}$$

They are integers. For example, this follows independently from the binomial expansions:

$$B_k = \binom{2k}{k} - \sum_{j=1}^{\lfloor k/44 \rfloor} \frac{\binom{2j}{j}}{2j-1}\, 4^{43j} \binom{2(k-44j)}{k-44j}. \tag{2}$$

Indeed $\binom{2j}{j}/(2j-1) = 2\,\mathrm{Cat}_{j-1}$ is an integer. This identity also fixes the branch and normalization unambiguously.

Define integers $C_0 = 1$, $C_1 = 99$ and

$$C_k = 198C_{k-1} - 10000C_{k-2} \quad (k \geq 2).$$

The cosine addition formula gives $C_k = 100^k \cos(kt)$. Finally put

$$A_0 = 1, \qquad A_k = 400A_{k-1} + B_k C_k.$$

Induction shows that

$$\frac{A_k}{400^k} = \sum_{j=0}^{k} \frac{B_j C_j}{400^j} = \mathcal{T}_k^{(43)}(t).$$

The following complete integer calculation proves

$$10000A_{1800} + 61 \cdot 400^{1800} > 0, \qquad -6099 \cdot 400^{1800} - 1000000A_{1800} > 0.$$

It uses only standard Python integer arithmetic. Every division in (1) is checked for zero remainder.

```
B = [1]
def at(k):
    return B[k] if k >= 0 else 0
for k in range(1, 1801):
    u = 4*(2*k-1)*at(k-1)
    u += 4**44*(2*k-132)*at(k-44)
    u += 4**45*(133-2*k)*at(k-45)
    v, r = divmod(u, 2*k)
    if r != 0:
        raise ArithmeticError(k)
    B.append(v)
C = [1, 99]
for k in range(2, 1801):
    C.append(198*C[-1] - 10000*C[-2])
A = 1
for k in range(1, 1801):
    A = 400*A + B[k]*C[k]
D = 400**1800
if not (10000*A + 61*D > 0
        and -6099*D - 1000000*A > 0):
    raise ArithmeticError("certificate failed")
print("Both strict bounds verified.")
```

Dividing by the positive denominator proves the theorem. □

Independent checks. The accompanying `verify_counterexample.py` additionally checks all 1801 coefficients against (2), all cosine values against the real parts of $(99 + i\sqrt{199})^k$, and the final sum in the reverse order. These checks use integers throughout. The scaled numerator has 15552 bits in absolute value; the denominator has 15559 bits. The saved certificate records both strict bounds and the exact numerator in hexadecimal.

Scope. This refutes Conjecture 1.3 and MathDB 369172. A negative real part at one boundary point does not exhibit a zero of the Taylor polynomial. The distinct zero-free Conjecture 1.2, recorded as MathDB 369171, therefore remains unresolved here. No minimality of d or n is claimed.

Task 2. Determine the parameter range for positivity

Classify the integers $d \geq 2$ for which $\mathcal{T}_n^{(d)}(t) > 0$ holds for every n and real t. In particular, settle the source's $d = 2$ positivity question.

Task 3. Determine the first failures

For each d admitting a negative cosine sum, determine the least truncation degree at which positivity fails and certify the relevant angles or intervals of angles.

Task 4. Prove a boundary scaling law

At each nontrivial $(d+1)$st root of unity, derive a uniform local asymptotic formula for the Taylor sections at angular distance of order $1/n$. Use it to identify parameter ranges with infinitely many negative cosine sums.

Task 5. Resolve the separate zero-free conjecture

Determine whether every Taylor section of $\sqrt{1 + z + \cdots + z^d}$, $d > 2$, is zero-free in the closed unit disk. An affirmative proof must accommodate the negative real parts proved here.

References

[1] L. V. Kovalev, *Sharp bounds for some segments of bounded power series*, arXiv:2507.04544v1 (2025), Conjectures 1.2 and 1.3, p. 2. https://arxiv.org/abs/2507.04544.